

ECN 594: Demographic Interactions and pyblp

Nicholas Vreugdenhil

Plan for today

1. **Recap: The IIA problem**
2. The demographic interaction model
3. Why demographics help with IIA
4. Introduction to pyblp
5. Worked example: Basic logit
6. Worked example: With demographics
7. Interpreting results

Recap: The IIA problem

- Last time: basic logit gives elasticities

$$\eta_{jj} = \alpha p_{jt}(1 - s_{jt}), \quad \eta_{jk} = -\alpha p_{kt}s_{kt}$$

- **Problem:** Cross-price elasticity η_{jk} doesn't depend on j !
- All products have the *same* cross-elasticity with product k
- Example: BMW raises price $\Rightarrow$ same fraction go to Mercedes as to Honda Civic
- This is unrealistic: BMW and Mercedes are closer substitutes
- **Today:** Fix this with demographic interactions

Plan for today

1. Recap: The IIA problem
2. **The demographic interaction model**
3. Why demographics help with IIA
4. Introduction to pyblp
5. Worked example: Basic logit
6. Worked example: With demographics
7. Interpreting results

The demographic interaction model

- We estimate a version of **Berry, Levinsohn & Pakes (1995)** — “BLP”
- Allow preferences to vary across consumers:

$$u_{ijt} = x_{jt}\beta_i + \alpha_i p_{jt} + \zeta_{jt} + \varepsilon_{ijt}$$

- β_i, α_i : consumer i 's taste for characteristics and price
- **Key question:** What drives heterogeneity in β_i and α_i ?
- **Our approach:** Observed demographics

$$\beta_{ik} = \beta_k + \sum_{\ell=1}^L \gamma_{k\ell} D_{i\ell}, \quad \alpha_i = \alpha + \sum_{\ell=1}^L \gamma_{\alpha\ell} D_{i\ell}$$

- $D_{i\ell}$: demographic ℓ for consumer i (income, age, family size, etc.)

Rewriting in terms of mean utility

- Substituting the expressions for β_{ik} and α_i :

$$u_{ijt} = \underbrace{x_{jt}\beta + \alpha p_{jt} + \zeta_{jt}}_{\delta_{jt} \text{ (mean utility)}} + \underbrace{\sum_{k,\ell} \gamma_{k\ell} D_{i\ell} x_{jt,k}}_{\mu_{ijt}} + \varepsilon_{ijt}$$

- **Mean utility** δ_{jt} : part common to all consumers
 - Contains “**linear parameters**” (β, α) — enter linearly in δ
- **Individual deviation** μ_{ijt} : how consumer i differs from average
 - Contains “**nonlinear parameters**” ($\gamma_{k\ell}$) — enter nonlinearly via shares
- Note: $\alpha < 0$ (higher price $\rightarrow$ lower utility)

From utility to demand

- **Individual choice probability** (with T1EV errors):

$$s_{ijt} = \frac{\exp(\delta_{jt} + \mu_{ijt})}{1 + \sum_{k=1}^J \exp(\delta_{kt} + \mu_{ikt})}$$

- **Market share** (aggregate over consumers):

$$s_{jt} = \frac{1}{I} \sum_{i=1}^I s_{ijt}$$

- This is **demand**: the fraction of consumers who buy product j
- Key: shares depend on mean utilities δ and demographics D_i

Examples of demographic interactions

- Recall: $\mu_{ijt} = \sum_{k,l} \gamma_{kl} D_{il} x_{jt,k}$ captures preference heterogeneity
- Effective price coefficient for consumer i : $\alpha_i = \alpha + \gamma_{\text{inc} \times p} \cdot \text{income}_i$
- **Question:** High-income consumers are typically *less* price-sensitive.
 - Given $\alpha < 0$, what sign should $\gamma_{\text{inc} \times p}$ have?

Where do demographics come from?

- Two scenarios:
 1. **Individual-level data:** Observe D_i for each consumer
 - Survey data, loyalty card data
 2. **Market-level data:** Know distribution of D_i in each market
 - Census data, Current Population Survey
 - This is more common in practice
- pyblp handles both cases

How do we estimate this model?

- Caveat: this is one algorithm, many variations exist
- **Two-step procedure:**
 1. **Estimate γ and δ_{jt} :**
 - Maximum likelihood with δ_{jt} as product fixed effects
 - Jointly estimates demographic coefficients γ
 2. **Recover β and α :**
 - Run 2SLS on $\delta_{jt} = x_{jt}\beta + \alpha p_{jt} + \zeta_{jt}$
 - Need instruments because price is endogenous
- **Key insight:** First step gives us δ_{jt} , then step 2 is linear regression with endogeneity (which we know how to handle!)

Plan for today

1. Recap: The IIA problem
2. The demographic interaction model
3. **Why demographics help with IIA**
4. Introduction to pyblp
5. Worked example: Basic logit
6. Worked example: With demographics
7. Interpreting results

Elasticities with demographic interactions

- **Elasticities with heterogeneous consumers:**

$$\eta_{jkt} = \begin{cases} \frac{p_{jt}}{s_{jt}} \sum_i \alpha_i s_{ijt} (1 - s_{ijt}) & \text{if } j = k \\ -\frac{p_{kt}}{s_{jt}} \sum_i \alpha_i s_{ijt} s_{ikt} & \text{otherwise} \end{cases}$$

- s_{ijt} : probability that consumer i purchases product j
- **Key insight:** Cross-elasticity now depends on $\sum_i s_{ijt} s_{ikt}$
 - Products that attract *similar consumers* have higher cross-elasticity
 - BMW and Mercedes attract similar (high-income) consumers $\Rightarrow$ closer substitutes
- Correlation in μ_{ijt} and μ_{ikt} drives realistic substitution patterns

Classic example: Nevo (2001) - Cereal demand

- “Measuring Market Power in the Ready-to-Eat Cereal Industry”
- Estimated demand for cereal using demographic interactions
- Key findings:
 - High-income households less price-sensitive
 - Families with children prefer kid-targeted cereals
 - Without demographics: wrong substitution patterns
- Result: Basic logit would underestimate markups!



Limitations: Demographics vs Mixed Logit

- **Demographics model** (what we use):
 - Heterogeneity only from observed characteristics
 - IIA still holds within demographic groups
 - Simple and tractable
- **Mixed logit / random coefficients** (advanced):
 - Heterogeneity from unobserved factors too
 - Fully relaxes IIA
 - Computationally intensive (requires simulation)
- For this course: demographics are sufficient
- Know that mixed logit exists for research

Plan for today

1. Recap: The IIA problem
2. The demographic interaction model
3. Why demographics help with IIA
4. **Introduction to pyblp**
5. Worked example: Basic logit
6. Worked example: With demographics
7. Interpreting results

What is pyblp?

- Python package for demand estimation
- Conlon & Gortmaker (2020), “Best Practices for Differentiated Products Demand Estimation with PyBLP”
- Handles:
 - Basic logit and logit with demographics
 - Instrumental variables
 - Standard errors
 - Post-estimation (elasticities, markups, etc.)

pyblp workflow

1. **Set up data:** products, markets, shares, characteristics
2. **Define formulation:** which variables, which IVs
3. **Create problem:** combine data and formulation
4. **Solve:** estimate the model
5. **Extract results:** coefficients, standard errors, elasticities

Let's walk through each step with car data

Step 1: Load and inspect data

- Load the BLP automobile data
- Key columns: market_ids, shares, prices, hpwt, air, mpd, space, demand_instruments0, ...

```
import pyblp
import pandas as pd

product_data = pd.read_csv(pyblp.data.BLP_PRODUCTS_LOCATION)
```

What data do we need?

- **Product data** (for each product j in market t):
 - Market share s_{jt}
 - Price p_{jt}
 - Characteristics x_{jt} (size, HP, fuel efficiency, etc.)
 - Instruments z_{jt} (BLP IVs, cost shifters)
- **Demographic data** (optional, for interactions):
 - Census data on income, family size, age by market
 - Or micro-level survey data

Plan for today

1. Recap: The IIA problem
2. The demographic interaction model
3. Why demographics help with IIA
4. Introduction to pyblp
5. **Worked example: Basic logit**
6. Worked example: With demographics
7. Interpreting results

The BLP automobile data

market_ids	shares	prices	hpwt	air	mpd	demand_instruments0	demand_instruments1
1971	0.0012	4.94	0.52	0	1.89	2.31	0.48
1971	0.0011	5.52	0.49	0	1.94	2.28	0.51
1971	0.0006	7.11	0.47	0	1.72	2.15	0.53
1971	0.0038	4.30	0.43	0	2.45	1.98	0.57
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

- 2,217 products across 20 years (1971–1990)
- BLP instruments: sums of rivals' characteristics

Model 1: Basic logit specification

- **Utility:**

$$u_{ijt} = \beta_1 \cdot \text{hpwt}_{jt} + \beta_2 \cdot \text{air}_{jt} + \beta_3 \cdot \text{mpd}_{jt} + \beta_4 \cdot \text{space}_{jt} + \alpha \cdot p_{jt} + \zeta_{jt} + \varepsilon_{ijt}$$

- **Parameters to estimate:**

- $\beta_1, \beta_2, \beta_3, \beta_4$: tastes for characteristics
- α : price sensitivity (expect $\alpha < 0$)

- **Note:** All consumers have the same preferences (no heterogeneity)

Estimating Model 1

- Define the formulation and solve

```
# Define formulation: characteristics + price (no constant)
formulation = pyblp.Formulation('0 + hpwt + air + mpd + space + prices
    ')

# Create problem (pyblp detects instruments automatically)
problem = pyblp.Problem(formulation, product_data)

# Estimate using 2SLS
results = problem.solve()
```

Model 1: Results

Variable	Coefficient	Std. Error
HP/Weight	-1.251	(0.623)
Air conditioning	1.773	(0.194)
Miles per dollar	-1.245	(0.047)
Space	-1.643	(0.097)
Price (α)	-0.225	(0.017)

- $\hat{\alpha} = -0.225 < 0$ (higher price $\rightarrow$ lower utility)
- Some coefficients have unexpected signs (omitted variable bias without constant)

Plan for today

1. Recap: The IIA problem
2. The demographic interaction model
3. Why demographics help with IIA
4. Introduction to pyblp
5. Worked example: Basic logit
6. **Worked example: With demographics**
7. Interpreting results

Agent data: Consumer demographics

- pyblp uses “agent data” for demographic draws:

market_ids	weights	income
1971	0.0005	1.10
1971	0.0007	0.45
1971	0.0005	1.27
1971	0.0007	0.23
⋮	⋮	⋮

- **weights**: Integration weight for each draw (sum to 1 per market)
- **income**: Income in \$100,000s (e.g., 1.10 = \$110k)
- pyblp integrates over these draws to compute market shares

Model 2: Adding demographic interactions

- **Utility with income interactions:**

$$u_{ijt} = \delta_{jt} + \underbrace{(\pi_{\text{inc}} \cdot \text{income}_i) \cdot p_{jt}}_{\mu_{ijt}} + \varepsilon_{ijt}$$

- Where $\delta_{jt} = \beta_1 \cdot \text{hpwt}_{jt} + \dots + \alpha_0 \cdot p_{jt} + \xi_{jt}$
- **New parameter:** π_{inc} (income $\times$ price)
 - If $\pi_{\text{inc}} > 0$: high-income less price-sensitive
- **Linear:** $\beta_1, \dots, \alpha_0$ (in δ_{jt}) **Nonlinear:** π_{inc} (in μ_{ijt})

Estimating Model 2

```
# Scale income to $100,000s for numerical stability
agent_data['income'] = agent_data['income'] / 100

X1 = pyblp.Formulation('0 + hpwt + air + mpd + space + prices')
X2 = pyblp.Formulation('0 + prices') # interacts with demographics
agent_formulation = pyblp.Formulation('0 + income')

problem = pyblp.Problem(
    (X1, X2), product_data, agent_formulation, agent_data
)
results = problem.solve(sigma=0, pi=np.array([[0.5]]))
```

Understanding the solve command

- `results = problem.solve(sigma=0, pi=np.array([[0.5]]))`
- **numpy** (`np`): Python library for numerical arrays
 - `np.array([[0.5]])` creates a matrix with value 0.5
- **sigma**: Standard deviations of random coefficients
 - `sigma=0` means **no random coefficients**
 - We use demographics only, not full mixed logit
 - (Mixed logit is beyond scope of this course)
- **pi**: Demographic interaction parameters (π)
 - Starting values for the optimizer (initial guess)
 - `pyblp` finds the best values via GMM
- **results**: Object containing all estimation output

Model 2: Results

Parameter	Estimate	Interpretation
<i>Linear parameters (in δ_{jt}):</i>		
α_0 (price)	-0.728	Base price sensitivity
<i>Demographic interaction (in μ_{ijt}):</i>		
π_{inc} (income $\times$ price)	+0.014	High-income less sensitive

- Effective price coef: $\alpha_0 + \pi_{\text{inc}} \cdot \text{income}_i$
- Income = \$50k: $-0.728 + 0.014 \times 0.5 = -0.721$
- Income = \$200k: $-0.728 + 0.014 \times 2.0 = -0.700$ (less price-sensitive)

Elasticity matrix example

	Prod 1	Prod 2	Prod 3	Prod 4	Prod 5
Prod 1	-3.5	0.02	0.03	0.01	0.02
Prod 2	0.02	-4.1	0.03	0.02	0.03
Prod 3	0.01	0.02	-2.8	0.01	0.02
Prod 4	0.02	0.03	0.02	-3.9	0.02
Prod 5	0.01	0.02	0.01	0.01	-3.2

- **Diagonal (red):** Own-price elasticities (negative, $|\eta| > 1$)
- **Off-diagonal:** Cross-price elasticities (positive for substitutes)
- Entry (j, k) : % change in s_j from 1% increase in p_k

Plan for today

1. Recap: The IIA problem
2. The demographic interaction model
3. Why demographics help with IIA
4. Introduction to pyblp
5. Worked example: Basic logit
6. Worked example: With demographics
7. **Interpreting results**

Post-estimation: Markups

- Recall: Lerner index $L = (p - MC)/p = 1/|\varepsilon|$
- Can recover markups from elasticities:

$$\text{markup}_j = \frac{p_j - mc_j}{p_j} = \frac{1}{|\eta_{jj}|}$$

- pyblp assumes Nash-Bertrand pricing
- Multi-product firms internalize substitution between own products

Worked example: Interpret this output

- pyblp gives you this output:

Variable	Estimate	Std. Error
Constant	3.5	0.8
HP/weight	1.2	0.4
Air cond.	0.9	0.3
Price	0.05	0.02

- **Question:** Something is wrong. What is it?

Take 1 minute to identify the problem.

Worked example: Interpret this output (solution)

- **Problem:** Price coefficient is positive (+0.05)!
 - Higher prices $\rightarrow$ higher utility. Economically nonsensical.
- **Why might this happen?**
 1. Weak instruments: IVs don't predict prices well
 2. Wrong signs: Maybe you forgot a negative sign somewhere
 3. Endogeneity still present: IVs are correlated with ξ
- **Fix:** Check first-stage F-stat, try different IVs, verify data

Common pyblp errors and how to fix them

- **“Singular matrix” error:**
 - Likely: collinear variables in formulation
 - Fix: Remove redundant variables
- **Price coefficient positive or near zero:**
 - Likely: weak instruments
 - Fix: Check first-stage F-stat; try different IVs
- **Huge standard errors:**
 - Likely: weak identification or too few observations
 - Fix: Aggregate to fewer markets; add more IVs

How to check your results make sense

1. **Price coefficient negative?** (Required!)
2. **Elasticities reasonable?**
 - Own-price: typically -2 to -10 for durables
 - Cross-price: positive but small
3. **Markups reasonable?**
 - Cars: typically 15-30%
 - Cereal: typically 30-50%
4. **First-stage F-stat > 10 ?** (IV relevance)
5. **Coefficients stable across specifications?**

This prepares you for HW1

- HW1 asks you to:
 1. Estimate basic logit (like Model 1)
 2. Estimate with demographics (like Model 2)
 3. Compute elasticities
 4. Interpret your results
- Key differences in HW1:
 - Uses cereal data (not cars)
 - Simpler: just sugar + price (not 5 characteristics)
 - Same structure: Model 1 $\rightarrow$ Model 2

Key Points

1. **Elasticities:** Own = $\alpha p_{jt}(1 - s_{jt})$; Cross = $-\alpha p_{kt}s_{kt}$ ($\alpha < 0$)
2. **IIA problem:** Cross-elasticities don't depend on product similarity
3. **Demographic interaction model:** $u_{ijt} = \delta_{jt} + \sum_{k,\ell} \gamma_{k\ell} D_{i\ell} x_{jt,k} + \varepsilon_{ijt}$
4. Demographics allow **heterogeneous preferences** and help with IIA
5. **pyblp workflow:** Data $\rightarrow$ Formulation $\rightarrow$ Problem $\rightarrow$ Solve $\rightarrow$ Interpret
6. **Key outputs:** Coefficients (check signs), standard errors, elasticities, markups
7. Price coefficient should be **negative**; OLS biases it toward zero
8. This is the foundation for HW1